

DevOps Deployment Details

This document captures the current container deployment details for the two images already built and pushed to ACR.

Registry Images

ACR login server:

``devengracr.azurecr.io``

Images:

- Gateway API: ``devengracr.azurecr.io/seabaasapigateway-core-api:latest``
- Developer Portal: ``devengracr.azurecr.io/seabaasapigateway-dev-portal:latest``

Both images were built for ``linux/amd64``.

Runtime Ports

Gateway API

- Container port: ``8080``
- Source: `[src/Gateway.Api/Dockerfile](../src/Gateway.Api/Dockerfile)`
- Docker runtime setting: ``ASPNETCORE_URLS=http://+:8080``
- Exposed port: ``8080``

Important note:

- Local development config in `[src/Gateway.Api/appsettings.Development.json](../src/Gateway.Api/appsettings.Development.json)` sets Kestrel to ``http://localhost:5001``
- Container deployments should use ``8080``, not ``5001``

Developer Portal

- Container port: ``3000``
- Source: `[src/gateway-portal-developer/Dockerfile](../src/gateway-portal-developer/Dockerfile)`
- Exposed port: ``3000``
- Startup command: ``npm run start``

Important note:

- Local dev mode normally uses ``3000``
- On one local run it shifted to ``3001`` only because ``3000`` was already occupied
- Container deployments should use ``3000``

Health and Probes

Gateway API

- Health endpoint: `GET /health`
- Source:
[src/Gateway.Api/Health/HealthCheckExtensions.cs](../src/Gateway.Api/Health/HealthCheckExtensions.cs)
- Suggested Kubernetes probes:
 - Liveness: `http://<pod-ip>:8080/health`
 - Readiness: `http://<pod-ip>:8080/health`

Developer Portal

- No dedicated health endpoint was found in the current app
- Suggested temporary readiness check: `GET /login` on port `3000`
- Suggested temporary liveness check: `GET /` or `GET /login` on port `3000`

Required Environment Configuration

Gateway API

The API throws on startup if these values are missing:

- `ConnectionStrings\_\_PostgreSQL`
- `ConnectionStrings\_\_Redis`
- `Kafka\_\_BootstrapServers`

Recommended explicit production settings:

- `ConnectionStrings\_\_PostgreSQL=<postgres connection string>`
- `ConnectionStrings\_\_Redis=<redis connection string>`
- `Kafka\_\_BootstrapServers=<host:port[,host:port]>`
- `Jwt\_\_SecretKey=<32+ char secret>`
- `Jwt\_\_Authority=<issuer url if used>`
- `Jwt\_\_Audience=SeaBaasAPIGateway`
- `Jwt\_\_RequireHttpsMetadata=true`
- `Jwt\_\_ValidateIssuer=true`
- `Jwt\_\_ValidateAudience=true`
- `OpenTelemetry\_\_ServiceName=SeaBaasAPIGateway`
- `OpenTelemetry\_\_ServiceVersion=1.0.0`
- `ASPNETCORE\_URLS=http://+:8080`

Operational note:

- [src/Gateway.Api/appsettings.json](../src/Gateway.Api/appsettings.json) still defaults `Jwt:Audience` to `gateway-api`
- [src/Gateway.Api/appsettings.Development.json](../src/Gateway.Api/appsettings.Development.json) uses `SeaBaasAPIGateway`

- DevOps should set `Jwt\_\_Audience` explicitly in the deployment to avoid ambiguity

Database note:

- The code defaults PostgreSQL search path to ``SeaBaasAPIGateway-Core`` when the connection string omits `Search Path`
- Source: [src/Gateway.Api/Configuration/ConfigurationExtensions.cs](../src/Gateway.Api/Configuration/ConfigurationExtensions.cs)

Database Scripts

This section lists all database SQL migrations and database-related helper scripts currently present in the repository.

All Migration SQL Files

Location: [database/migrations](../database/migrations)

- [database/migrations/000_create_schema.sql](../database/migrations/000_create_schema.sql)
- [database/migrations/001_initial_schema.sql](../database/migrations/001_initial_schema.sql)
- [database/migrations/002_seed_data.sql](../database/migrations/002_seed_data.sql)
- [database/migrations/003_add_login_endpoint.sql](../database/migrations/003_add_login_endpoint.sql)
- [database/migrations/004_load_balancing.sql](../database/migrations/004_load_balancing.sql)
- [database/migrations/005_convert_ids_to_uuid.sql](../database/migrations/005_convert_ids_to_uuid.sql)
- [database/migrations/006_enterprise_identity.sql](../database/migrations/006_enterprise_identity.sql)
- [database/migrations/007_org_access_management.sql](../database/migrations/007_org_access_management.sql)
- [database/migrations/008_import_seabaas_smartadapter.sql](../database/migrations/008_import_seabaas_smartadapter.sql)
- [database/migrations/009_dev_destinations.sql](../database/migrations/009_dev_destinations.sql)

Recommended full execution order for a fresh environment:

- 1. 000_create_schema.sql
- 2. 001_initial_schema.sql
- 3. 002_seed_data.sql
- 4. 003_add_login_endpoint.sql
- 5. 004_load_balancing.sql
- 6. 005_convert_ids_to_uuid.sql
- 7. 006_enterprise_identity.sql
- 8. 007_org_access_management.sql
- 9. 008_import_seabaas_smartadapter.sql
- 10. 009_dev_destinations.sql

All Database-Related Scripts

Location: [scripts](../scripts)

- [scripts/setup-db.sh](../scripts/setup-db.sh)

Purpose: initializes database and applies early schema and seed scripts (001 and 002).

- `[scripts/migrate-db.sh](../scripts/migrate-db.sh)`

Purpose: applies the scripted migration sequence from 001 through 007.

- `[scripts/apply-login-migration.sh](../scripts/apply-login-migration.sh)`

Purpose: applies only 003_add_login_endpoint.sql.

- `[scripts/import_swagger_seed.py](../scripts/import_swagger_seed.py)`

Purpose: imports swagger-driven seed data.

- `[scripts/test-login.sh](../scripts/test-login.sh)`

Purpose: validates login behavior after migration.

- `[scripts/test-admin-persistence.sh](../scripts/test-admin-persistence.sh)`

Purpose: validates admin persistence behavior.

- `[scripts/test_e2e.py](../scripts/test_e2e.py)`

Purpose: end-to-end verification.

- `[scripts/latency_test.py](../scripts/latency_test.py)`

Purpose: latency/performance verification support.

Deployment Notes for DevOps

- `[scripts/migrate-db.sh](../scripts/migrate-db.sh)` does not currently include 000, 008, or 009.
- For complete deployment parity, apply 000 first and then apply 008 and 009 after running migrate-db.sh.
- Ensure DB connection environment variables are set before execution:

- DB_HOST
- DB_PORT
- DB_NAME
- DB_USER
- DB_PASSWORD

Developer Portal

Public runtime environment variables:

- ``NEXT_PUBLIC_GATEWAY_BASE_URL``
- ``NEXT_PUBLIC_PORTAL_NAME``
- ``NEXT_PUBLIC_SUPPORT_EMAIL``

Current defaults from `[src/gateway-portal-developer/lib/config/env.ts](../src/gateway-portal-developer/lib/config/env.ts)`:

- ``NEXT_PUBLIC_GATEWAY_BASE_URL``=` empty by default
- ``NEXT_PUBLIC_PORTAL_NAME``=SeaBaas Developer Portal`
- ``NEXT_PUBLIC_SUPPORT_EMAIL``=support@gapeiro.dev`

Recommended deployment values:

- ``NEXT_PUBLIC_GATEWAY_BASE_URL=https://<gateway-public-base-url>``
- ``NEXT_PUBLIC_PORTAL_NAME=SeaBaas Developer Portal``
- ``NEXT_PUBLIC_SUPPORT_EMAIL=support@gapeiro.dev``
- ``PORT=3000``

OAuth provider variables are optional and only needed if social login is enabled:

- ``GITHUB_CLIENT_ID``
- ``GITHUB_CLIENT_SECRET``
- ``GOOGLE_CLIENT_ID``
- ``GOOGLE_CLIENT_SECRET``
- ``AZURE_AD_CLIENT_ID``
- ``AZURE_AD_CLIENT_SECRET``
- ``AZURE_AD_TENANT_ID``
- ``APPLE_CLIENT_ID``
- ``APPLE_CLIENT_SECRET``

Auth note:

- If OAuth env vars are missing, the NextAuth route returns ``500`` for OAuth sign-in attempts
- The mocked email/password login route under `[src/gateway-portal-developer/app/api/auth/login/route.ts](../src/gateway-portal-developer/app/api/auth/login/route.ts)` still returns a successful mock session

Example Container Settings

Gateway API

- Image: ``devengracr.azurecr.io/seabaasapigateway-core-api:latest``
- Container port: ``8080``
- Probe path: ``/health``

Developer Portal

- Image: ``devengracr.azurecr.io/seabaasapigateway-dev-portal:latest``
- Container port: ``3000``
- Probe path: ``/login`` until a dedicated health endpoint is added

Summary For DevOps

- Gateway API image: ``devengracr.azurecr.io/seabaasapigateway-core-api:latest``
- Gateway API port: ``8080``
- Gateway API health check: ``/health``
- Developer Portal image: ``devengracr.azurecr.io/seabaasapigateway-dev-portal:latest``
- Developer Portal port: ``3000``
- Developer Portal temporary health check: ``/login``